Portfolio Milestone Module 3: Neural Networks for CubeSat Telemetry Anomaly Detection

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CSC510: Foundations of Artificial Intelligence

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August 9, 2026

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The “Portfolio Milestone Module 2: Use-Case Scenario Proposal” (Ricciardi, 2026) proposes using a combination of an Artificial Neural Network (ANN) and an A* symbolic planner for a CubeSat telemetry anomaly-detection system. Within this hybrid approach, the ANN will handle OPS-SAT telemetry segment classification by labeling them as having normal or anomalous behavior, while the A* symbolic planner will organize diagnostic actions into an investigation sequence, and a similarity search will retrieve historical telemetry cases. The process of labeling the telemetry segment is defined “fuzzy” tasks because it requires recognizing uncertain, noisy, and nonlinear patterns within sensor measurements, and ANNs are well suited for such tasks (Ruszczak et al., 2025). This paper defines the learning method and initial architecture of the neural network component of the hybrid CubeSat telemetry anomaly-detection system. The ANN will integrate Supervised Learning (SL) within a binary-classification process using the labeled OPSSAT-AD data. The available OPSSAT-AD dataset contains telemetry segment data labeled as normal or anomalous that can be used as SL targets during training. The ANN model will benefit most from having a small deep feedforward network architecture implemented as a fully connected Multilayer Perceptron (MLP) with three hidden layers containing 32, 16, and 8 neurons. This architecture should provide the needed depth to learn nonlinear relationships from the telemetry data while being small enough to prevent overfitting. During supervised training, the model will compare its classification prediction against the labels from the OPSSAT-AD dataset. This will allow the model to adjust its weights and biases through backpropagation by reducing the error between the predicted and target outputs (Sharda et al., 2024). Additionally, a Rectified Linear Unit (ReLU) activation function will be implemented within the hidden layers to learn nonlinear relationships, and a sigmoid activation function in the output layer will be used to generate anomaly probabilities between 0 and 1. Finally, the A* symbolic planner will use the model ANN outputs, together with the system’s diagnostic rules, to organize diagnostic actions into an investigation sequence.

References

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